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Bridging Worlds: CloudNativePG as the Serving Layer for Modern Lakehouse Architectures

Gabriele Quaresima,
Lakehouse Day, Glasgow,
10/10/2026



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- <3 OSS
- <3 Coding
- ♫ Playing sax
- <3 Paola and Maria

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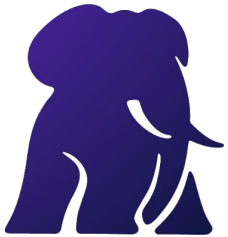
Survey

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CloudNativePG

- **CloudNativePG** is a CNCF Sandbox project. It is an open-source operator designed to manage **PostgreSQL** workloads on any supported **Kubernetes** cluster

- It defines a new K8s resource called **Cluster**, representing a PostgreSQL cluster made up of a single primary and an optional number of replicas



CloudNativePG

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PostgreSQL

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PostgreSQL

PostgreSQL is an object-relational **database** management system.

The first PostgreSQL release formed version 6.0 on January 1997.

Latest stable release is 18.6 (August 2026), 19 Beta is out.

Because of the **liberal license**, PostgreSQL can be used, modified, and distributed by anyone free of charge for any purpose, be it private, commercial, or academic.

Full official documentation: postgresql.org/docs/current

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PostgreSQL - Features

It supports a large part of the SQL standard and offers many modern features:

- Complex queries
- Foreign keys
- Triggers
- Updatable views
- Transactional integrity
- Multiversion Concurrency Control

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PostgreSQL - Extensibility

PostgreSQL can be extended by the user in many ways, for example by adding new:

- Data types
- Functions
- Operators
- Aggregate functions
- Index methods
- Procedural languages

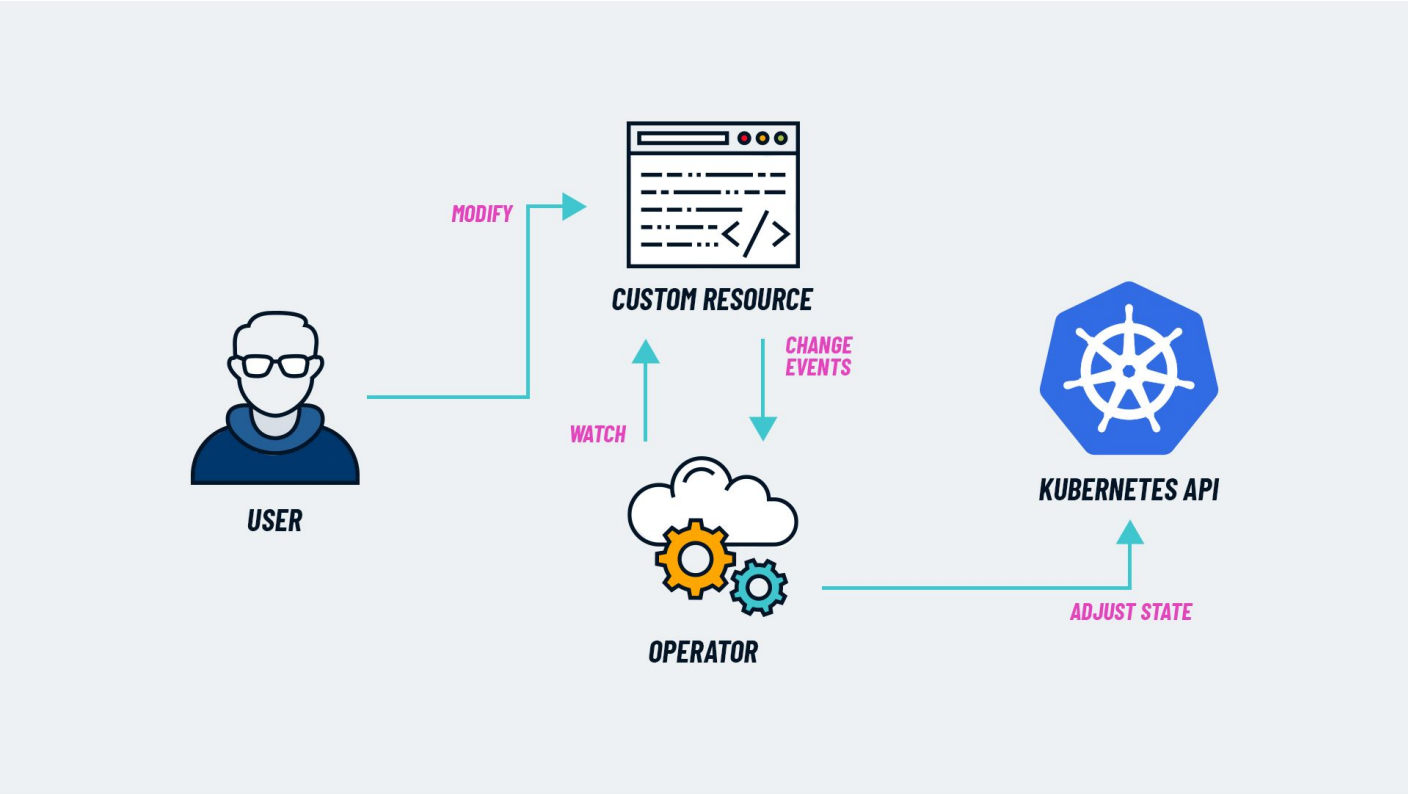
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Kubernetes operators

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Kubernetes Operator



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Kubernetes Operator - In a nutshell

- Software extensions to Kubernetes
- Use of **custom resources** to manage applications and their components
- Operators follow Kubernetes principles, notably the control loop
- Operators implement and automate common Day-1 (installation, configuration, etc) and Day-2 (re-configuration, update, backup, failover, restore, etc.) activities
- Via operators, you can stop treating an application as a collection of primitives like Pods, Deployments, Services or ConfigMaps, but instead as a single object that only exposes the knobs that make sense for the application

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Bring Kubernetes to Postgres, bring Postgres to Kubernetes!



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Deep into CloudNativePG

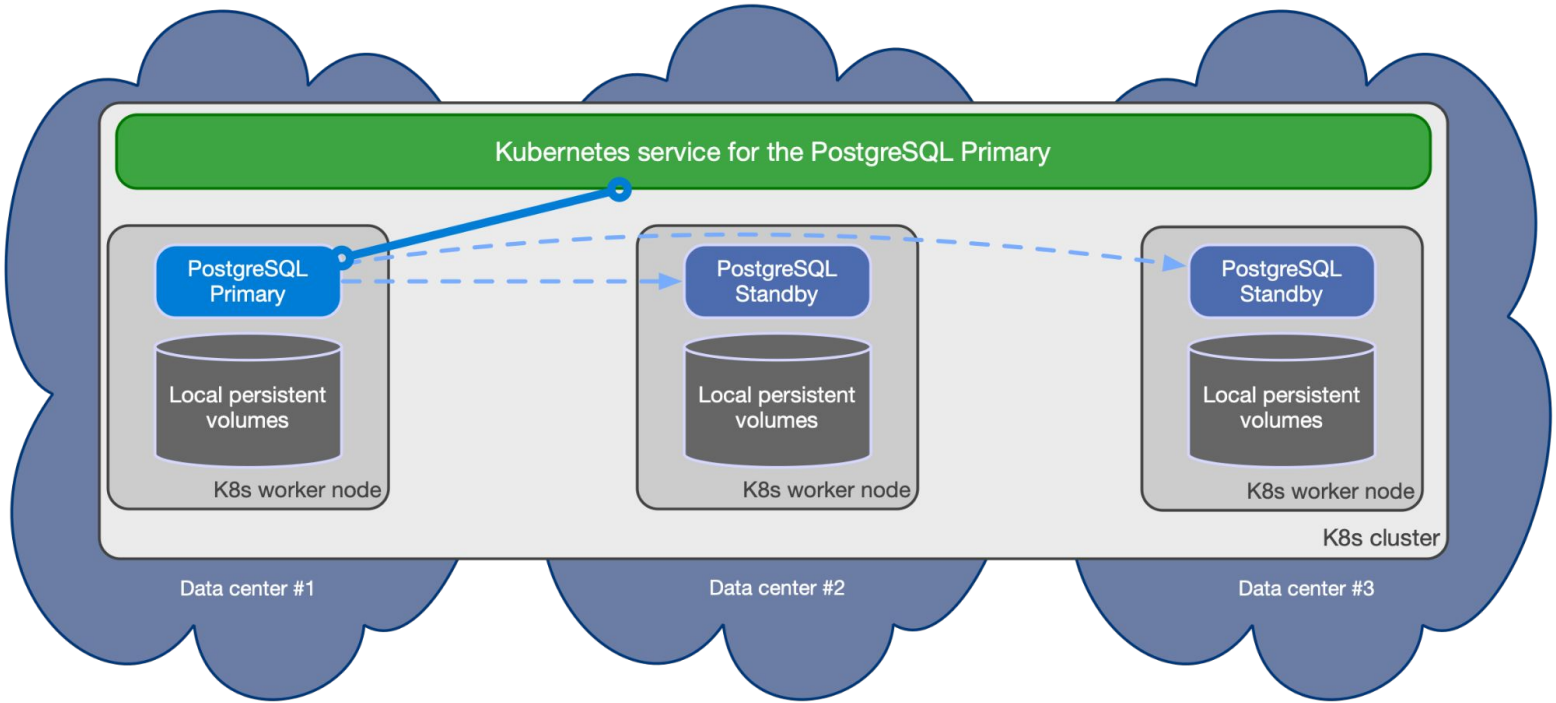
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CloudNativePG - Cluster CRD example

```
> batcat cloudnative-pg/docs/src/samples/cluster-example.yaml
       File: cloudnative-pg/docs/src/samples/cluster-example.yaml
   1   apiVersion: postgresql.cnpg.io/v1
   2   kind: Cluster
   3   metadata:
   4     name: cluster-example
   5   spec:
   6     instances: 3
   7
   8     storage:
   9       size: 1Gi
```

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CloudNativePG - Suggested architecture



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CloudNativePG - Main features

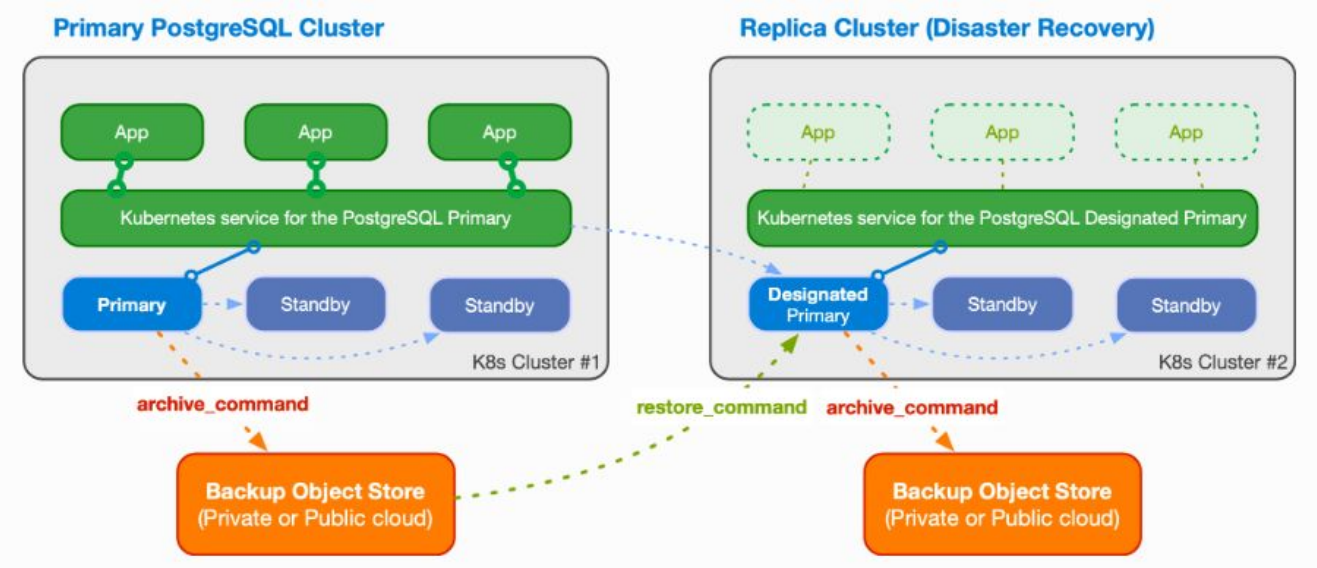
- Direct integration with K8s Api Server for self healing
- High Availability capability
- Planned switchover
- Scale up/down capabilities
- Definition of an arbitrary number of instances, rw and ro services
- Declarative management of PostgreSQL configuration
- Support for local PV with PVC templates
- Backup on VolumeSnapshot and object stores
- Major PG upgrade
- kubectl plugin for cluster management
- Declarative tablespaces, databases, publications and subscriptions
- CNPG-I: define your own plugin!

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CloudNativePG - Deployment across Kubernetes clusters

Distributed PostgreSQL topology with **replica clusters** enables multi-cluster deployments in private, public, hybrid, and multi-cloud contexts.

A replica cluster is a separate **Cluster** resource that is in continuous recovery, replicating from another source, either via WAL or streaming replication.



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CloudNativePG - Operand images

- The PostgreSQL operand **container images** are available for all PGDG supported versions of PostgreSQL, across multiple architectures, directly from the postgres-containers project's GitHub Container Registry.

- Daily jobs ensure that CVEs in the entire stack are promptly addressed

- Additionally, the community provides images for the PostGIS extension

- Check this blog article, if you want create your own **custom** container image.

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CloudNativePG - Observability

- CloudnativePG exposes native **Prometheus** exporters. Some are already defined, but you can extend them and add new ones

- All logs are in stdout
  and in **JSON** format

- CNPG plugin **status** command



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CloudNativePG - Kubectl Plugin

CloudnativePG provides a **plugin** for kubectl to manage a cluster in Kubernetes

```
> ./kubectl-cnpg --help
A plugin to manage your CloudNativePG clusters

Usage:
  kubectl cnpg [command]

Operator-level administration
  install      CloudNativePG installation-related commands

Troubleshooting
  logs         Logging utilities
  report       Report on the operator or a cluster for troubleshooting

Cluster administration
  destroy      Destroy the instance named [cluster]-[node] or [node] with the associated PVC
  fencing      Fencing related commands
  hibernate    Hibernation related commands
  maintenance  Sets or removes maintenance mode from clusters
  promote      Promote the pod named [cluster]-[node] or [node] to primary
  reload       Reload a cluster
  restart      Restart a cluster or a single instance in a cluster

Database administration
  backup       Request an on-demand backup for a PostgreSQL Cluster
  certificate  Create a client certificate to connect to PostgreSQL using TLS and Certificate authentication
  publication  Logical publication management commands
  snapshot     DEPRECATED (use `backup -m volumeSnapshot` instead)
  status       Get the status of a PostgreSQL cluster
  subscription Logical subscription management commands

Miscellaneous
  fio          Creates a fio deployment, pvc and configmap
  pgadmin4     Creates a pgAdmin deployment
  pgbench      Creates a pgbench job
  psql         Start a psql session targeting a CloudNativePG cluster

Additional Commands:
  completion   Generate the autocompletion script for the specified shell
  help         Help about any command
  version      Prints version, commit sha and date of the build
```

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Bridge to Lakehouse world

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CloudNativePG - Shared Object Storage

Delta Lake, **Apache Iceberg** and **Apache Hudi** store tables as plain files on S3-compatible object storage

- CloudNativePG could archive **WAL** and **backups** into S3, via the **barman-cloud plugin** (see **ObjectStore** CRD) or **Klio** tier-2
- Result: **one bucket, one policy set**: the same encryption, IAM, and retention rules for both
- No separate backup-vendor and data-lake-vendor security model to reconcile

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Object Storage in practice

- One bucket, two prefixes: **cnpg-backups/** from CloudNativePG, **warehouse/** from Spark/Flink batch jobs
- The **ObjectStore** CRD declares the destination once; any Cluster can reuse it
- Storage becomes the durable source of truth: compute can be rescheduled without touching data
- **Public lab:** RustFS holds cnpg-backups/ and warehouse/ side by side

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CloudNativePG - Foreign Data Wrappers

PostgreSQL's **Foreign Data Wrapper** framework lets a query reach outside the database: the planner treats it like a local table

- **pg_duckdb**, parquet_s3_fdw, and pg_analytics embed an engine that reads Parquet and Lakehouse formats natively
- Result: query Lakehouse tables directly: **no nightly ETL, no duplicated copy**
- The payoff: **JOIN** hot operational data with cold Lakehouse history in one query

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Foreign Data Wrappers in practice

- **pg_duckdb** embeds DuckDB inside Postgres: DuckDB scans, Postgres's planner joins
- Reader and writer only share a **storage location**, not a metastore
- Use cases: ad-hoc analytics, historical context in dashboards, live-vs-batch checks
- Trade-off: an analytical path, not the hot transactional one

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CloudNativePG - pgvector for Embedding Retrieval

- Spark/Flink turn unstructured data into **embeddings** and land them as Parquet/Iceberg
- **pgvector** adds a native vector type and distance operators, enabled via the Database CRD
- An **HNSW** (Hierarchical Navigable Small World) index turns similarity search into a sub-millisecond query
- The last mile: the Lakehouse computes embeddings in batch, CloudNativePG serves them live

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CloudNativePG - pgvector in practice

- Pipeline: unstructured data → embeddings → Parquet → pgvector → HNSW → millisecond query
- One engine serves vector search **and** relational filters together
- This is the **RAG retrieval layer**: CloudNativePG in front of an LLM
- Refresh model: batch re-embeds; Postgres just re-runs the load
- In the lab, **duckdb** stands in for Spark/Flink, writing the embeddings straight to Parquet on RustFS

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How to use CloudNativePG

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CloudNativePG - Installation

```
➜  ~ kubectl apply --server-side -f https://raw.githubusercontent.co
m/cloudnative-pg/cloudnative-pg/main/releases/cnpg-1.30.1.yaml
```

Refer to the CloudNativePG documentation for Helm Chart and Operator Lifecycle Manager (OLM), directly from OperatorHub.io

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CloudNativePG - Product

- CloudNativePG is **production ready**!

- CloudNativePG became the de facto standard to deploy **PostgreSQL in Kubernetes**

- Community is very active inside the Cloud Native Computing Foundation channels: **cloudnativepg-dev** and **cloudnativepg-users**

- CloudNativePG has been accepted into the **CNCF Sandbox**

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CloudNativePG - License

CloudNativePG has been originally created **EDB**

CloudNativePG is 100% **open source** and community-driven

All components are available under the Apache 2 license on Github

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CloudNativePG - Adopters

- **Google Cloud**
  https://cloud.google.com/kubernetes-engine/docs/tutorials/stateful-workloads/cloudnativepg

- **Microsoft Azure**
  https://learn.microsoft.com/en-us/azure/aks/postgresql-ha-overview

- **IBM**

- **EDB**

  EDB CloudNativePG Cluster, EDB CloudNativePG Global Cluster, EDB Postgres AI Cloud Service (DBaaS)

  Full list here

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CloudNativePG - Useful links

Website: https://cloudnative-pg.io/

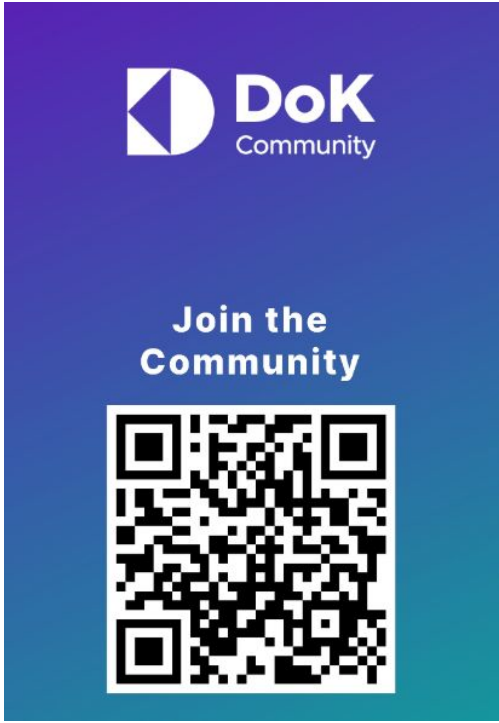
CloudNativePG is on LinkedIn, GitHub, cncf.io/sandbox-projects/ , but also on

- **Slack https://cloud-native.slack.com/archives/C08MAUJ7NPM**
- **Twitter https://twitter.com/CloudNativePg**
- **Mastodon https://mastodon.social/@CloudNativePG**
- **Bluesky https://bsky.app/profile/cloudnativepg.bsky.social**

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CloudNativePG - Resources

- CloudNativePG website and GitHub project
- CloudNativePG Recipes on Gabriele Bartolini's blog
- CloudNativePG maintainers and contributors are active on Data On Kubernetes community
- CNCF blog article "Cloud Neutral Postgres Databases with Kubernetes and CloudNativePG"
- Whitepaper with a benchmark of a PostgreSQL database in Kubernetes with CloudNativePG "EDP Postgres AI on OpenShift cluster using CSI driver for Dell PowerFlex"



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Try the lab!

https://github.com/gabriele-wolfox/lakehouse-day



Get in touch with me on

- #cloudnativepg-dev on CNCF Slack
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THANK YOU!

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